18 March 2026

Date: / /

Exercise 4
Join, Union, Aggregate

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School: BrightLearn

Question 01
SQL syntax

```
Select  employee_ID, first_name, last_name,
        Department, Salary, Project_ID, Project_name,
        Budget, Status
FROM    employees AS A
Inner Join  Projects AS B
ON  A.employee_ID = B.employee_ID
```

mployee_ID	First_name	last_name	Dept.	Salary	Project_ID	Project_name	Budget	status
1	John	Doe	IT	70000	101	AI dev	100000	completed
1	John	Doe	IT	70000	103	Cyber Aud	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Tr.	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial An.	90000	On Hold
5	Emma	Wilson	Sales	65000	105	Market Exp.	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Manag	80000	Pending

Question 02

Today's mood:
- Craving coffee
- 11/10
- Logging off
- Main character energy
- BRB overthinking
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Question 02

SQL Syntax

```
Select Employee_ID, First_name, last_name, Department,
       Salary, Project_ID, Project_name, Budget,
       Status
FROM employees AS A
Left Join Projects AS B
ON A.employee_ID = B.employee_ID;
```

OUTPUT Table

employee ID	First_name	last_name	Dept.	Salary	Project-ID	Project name	Budget	Status
1	John	Doe	IT	70000	101	AI Dev	100000	Completed
1	John	Doe	IT	70000	103	Cyber	75000	Pending
2	Alice	Smith	HR	60000	102	Employee	50000	On Going
3	Bob	Johnson	Finance	75000	104	FinanceA	90000	On Going
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market ext.	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Man	80000	Pending

Question 04

SQL Syntax

```
Select employee_id, First_name, last_name,
       Department, Salary, Project_ID, Project_name,
       Budget, status
FROM employees AS A
Full Outer Join Projects AS B
ON A.employee_ID = B.employee_ID;
```

OUTPUT table same as Question 02

Today's mood:

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~~Question 7~~
Question 3
filtering statements
37.

SQL Syntax
Select employee_ID,
First_name,
last_name,
Department,
Salary
FROM employees
WHERE Salary > 70 000;

Output table

Employee ID	First name	last name	Dept.	salary
3	Bob	Johnson	Finance	15000
4	David	Brown	IT	12000
6	Micheal	Clark	Finance	80000

38. SQL Syntax
Select employee_ID,
First_name,
last_name,
Department,
Salary
FROM employees
WHERE Department IN('IT', 'Finance');

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Output Table

Date: / /

employee ID	First_name	last_name	Department	Salary
1	John	Doe	IT	10 000
3	Bob	Johnson	Finance	15 000
4	David	Brown	IT	12 000
6	Michael	Clark	Finance	80 000

3.9. SQL syntax

```
Select  Project_ID,
        Project_name,
        Budget,
        Status
FROM    Projects
WHERE   Status = 'completed';
```

Output table

Project_ID	Project_name	Budget	Status
102	Employee Train.	50 000	On Going
103	Cybersecurity	15 000	Pending
104	Financial Analysis	90 000	On Going
106	Risk Management	80 000	Pending

3.10.

```
Select  Project_ID,
        Project_Name,
        Budget,
        Status
FRom    Projects
WHERE   Buget 7
```

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Buget >70 000 AND status <> 'completed',

OUT PUT Table

Project_ ~~name~~ID	Project_ name	Budget	status
103	Cyberfeurity Audit	75000	Pending
104	Financial Analysis	90 000	On Going
105	80 000	80 000	Pending

3.11. SQL Syntax

```
Select  employee_ID,
        First_name,
        last_name,
        Department,
        salary,
        city
FROM    employees
WHERE   City IN ('New York', Toronto')
ORDER BY Salary DESC;
```

OUT PUT Table

employee ID	First name	last name	Dept	Salary	city
6	Micheal	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

~~Question~~

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3.12. SQL Syntax

```
Select employee_ID,
       First_name,
       last_name,
       Department,
       Salary
FROM employees
ORDER BY Salary DESC;
```

OUTPUT Table

employee_ID	First_name	last_name	Dept	Salary
6	Micheal	clark	Finance	80 000
3	Bob	Johnson	Finance	75000
4	David.	Brown	IT	72 000

Question 4. Aggregate functions, Group B

4.13.

```
Select Department
       Sum(Salary) As Total_Salary
FROM employee
Group by Department
ORDER BY Total_Salary DESC;
```

Department	Total_Salary
Finance	155 000
IT	142 000
Sales	65 000
HR	60 000

OUT-PUT Table

Today's mood:

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4.14. SQL Syntax

```
Select city,
       AVG(Salary) AS Average_Salary
FROM employees
~~Where Average_Salary > 65000~~
Group BY city
Having AVG(Salary) > 6500 0;
```

OUTPUT Table

City	Average_Salary
New York	75000
Toronto	75000
London	72000

4.15 SQL Syntax

```
Select department,
       count(employee_id) AS Employee_Count
FROM employees
Group BY department
Having Count(employee_id) > 1;
```

OUTPUT

Department	Employee_Count
IT	2
Finance	2

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4.16. SQL Syntax

```
Select count(Project ID) AS Project_count
       status
FROM Projects
Group By status
Having COUNT(Project_ID) >= 2;
```

Output table

Status	Project_Count
On Going	2
Pending	2

4.17. SQL Syntax

```
Select employee_ID,
       First_name,
       last_name
       Sum(Budget) AS Total Project Budget
FROM employees AS A
Inner Join Projects AS B
ON A.employee_ID = B.employee_ID
Group By employee ID, first_name, last_name
Having Sum(Budget) > 150 000;
```

Output Table

Emp. ID,	First name	lastname	Total Project Budget
1	John	Doe	175 000

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